

Classification Methods for Diabetes Prediction

Pattern Recognition and Machine Learning Project

June 2026

1 Exploratory Data Analysis (EDA)

1.1 Introduction and Problem Formulation

Diabetes mellitus is a chronic metabolic disease characterized by persistently elevated blood glucose levels resulting from impaired insulin production, insulin action, or both. It is one of the most common chronic diseases worldwide and represents a major public health challenge.

Several risk factors contribute to the development of diabetes, including obesity, physical inactivity, unhealthy diet, genetic predisposition, advanced age, hypertension, and a family history of the disease. Elevated glucose levels, increased body mass index (BMI), and abnormal insulin regulation are among the most important clinical indicators associated with diabetes.

If left untreated, diabetes can lead to severe long-term complications affecting multiple organs and body systems. Common complications include cardiovascular diseases, kidney failure, nerve damage, vision impairment, and an increased risk of stroke. These complications significantly reduce quality of life and may increase mortality.

Early detection of diabetes is therefore essential. Identifying high-risk individuals before severe complications develop allows for timely medical intervention, lifestyle modifications, and appropriate treatment. Machine learning methods can assist healthcare professionals by analyzing patient data and identifying patterns associated with diabetes, enabling faster and more accurate diagnosis.

In this project, machine learning classification techniques are applied to predict whether a patient has diabetes based on several medical measurements. The analysis is performed using the **Pima Indians Diabetes Dataset**, a widely used benchmark dataset for diabetes prediction available on Kaggle, where the target variable is Outcome (1 = diabetes, 0 = no diabetes). Before applying classification algorithms, a comprehensive Exploratory Data Analysis (EDA) and data preprocessing procedure are performed to identify missing values, duplicate records, outliers, and relationships between features. The goal is to implement and compare three baseline classification methods — Gaussian Naive Bayes, QDA, and KNN — alongside two extensions of KNN (Distance-Weighted KNN and Feature-Weighted KNN), all implemented from scratch, and to evaluate which method best predicts diabetes on this dataset.

1.2 Data Cleaning and Preprocessing

1.2.1 Dataset Structure and Shape

The dataset contains 768 rows (observations) and 9 columns. It consists of 8 numerical medical features and 1 target variable (**Outcome**). The features describe attributes such as **Glucose** levels, **BloodPressure**, **SkinThickness**, **Insulin**, **BMI**, **Pregnancies**, **DiabetesPedigreeFunction** (an index of hereditary diabetes risk based on family history), and **Age**.

1.2.2 Missing Values and Duplicates

We see that there are no duplicate rows, meaning every patient record is unique. Furthermore, there are no standard missing (NaN) values in the dataset.

However, looking closely at the minimum values, we noticed that **Glucose**, **BloodPressure**, **SkinThickness**, **Insulin**, and **BMI** have a minimum value of 0.0. Since these physiological measurements cannot be zero for a living person, these zeros actually represent missing data.

1.2.3 Imputation Strategy

To handle these missing values without dropping records, we used a **per-class median imputation**. Instead of a simple global average, we replaced the zeros with the median value of that specific feature, calculated separately for each **Outcome** class (0 or 1). This approach preserves the natural statistical differences between the two groups.

1.2.4 Feature Distributions

After imputation, we analyzed the distributions of the numerical features. Variables like **Glucose** and **BloodPressure** follow an approximately normal (Gaussian) distribution. This makes them well-suited for methods like Naive Bayes and QDA.

In contrast, features such as **Pregnancies**, **Age**, **Insulin**, and **DiabetesPedigreeFunction** are heavily right-skewed.

1.2.5 Feature Comparison Between Classes

To see how medical measurements change based on the diagnosis, we compared the feature values between the two classes (**Outcome** 0 and 1). Using boxplots, we observed clear visual differences:

- Patients with diabetes (**Outcome** = 1) show noticeably higher median levels of **Glucose**, **Insulin**, and **BMI** compared to healthy individuals.
- Features like **BloodPressure** and **Age** also display upward shifts in the diabetic group, though their distributions overlap more between classes.

This step confirms that these physiological variables carry strong discriminative information to help our classification models.

1.2.6 Class Distribution and Feature Correlations

Finally, we analyzed the overall distribution of the target variable and the correlation matrix. We found that 34.9% of the patients have diabetes (1) and 65.1% do not (0), which shows that the dataset is imbalanced.

Additionally, the Pearson correlation matrix mathematically verifies our boxplot observations. **Glucose**, **Insulin**, and **BMI** display the strongest positive correlation coefficients with the **Outcome**.

The final, preprocessed dataset maintains its original structural shape of 768×9 and is ready for model training.

2 Data Preparation

After the exploratory analysis and preprocessing steps described in the previous section, the dataset is ready for classification. Before training the machine learning models, the input features and target variable are separated, the dataset is divided into training and testing

subsets, and feature standardization is applied. These steps ensure that the models are trained and evaluated under consistent conditions.

2.1 Train-Test Split

The dataset is divided into input features and the target variable. The target variable is `Outcome`, where a value of 1 indicates diabetes and a value of 0 indicates no diabetes. A manual train-test split is performed, assigning 80% of the observations to the training set and 20% to the test set. Prior to splitting, the data is randomly shuffled to ensure a representative distribution of samples in both subsets.

The resulting training set contains 615 observations, while the test set contains 153 observations.

2.2 Feature Standardization

The medical measurements in the dataset are expressed on different scales. For example, Glucose is measured in mg/dL, Age is measured in years, and BMI is expressed in kg/m². To prevent features with larger numerical values from dominating the learning process, standardization is applied.

The mean and standard deviation are computed using only the training set. These statistics are then used to transform both the training and test data according to

$$x_{scaled} = \frac{x - \mu}{\sigma},$$

where μ denotes the feature mean and σ denotes the feature standard deviation calculated from the training set.

Using only training-set statistics prevents information leakage from the test set and ensures a fair evaluation of model performance.

3 Gaussian Naive Bayes Classification

Gaussian Naive Bayes was selected as the first baseline classification method because it is simple, fast, and directly connected to probabilistic decision theory. The method estimates the probability that a patient belongs to each class and assigns the patient to the class with the highest posterior probability.

In this project, the two possible classes are `Outcome = 0`, representing patients without diabetes, and `Outcome = 1`, representing patients with diabetes. Since all input variables are numerical medical measurements, the Gaussian version of Naive Bayes is used.

3.1 Model Implementation

The Gaussian Naive Bayes classifier was implemented from scratch using NumPy. For each class, the algorithm first extracts the corresponding training samples and estimates the class prior probability. It then computes the mean and variance of every feature, which serve as the parameters of the Gaussian distributions.

To classify a new observation, the logarithm of the Gaussian probability density function is used to calculate class likelihoods. The posterior probabilities are obtained by combining these likelihoods with the class priors, and the observation is assigned to the class with the highest posterior probability.

3.2 Model Training and Prediction

After estimating the model parameters, the Gaussian Naive Bayes classifier was trained using the prepared training dataset. The learned parameters consisted of class prior probabilities together with the mean and variance of each feature within both classes.

The trained model was then applied to the test set containing 153 previously unseen patient records. For each patient, the posterior probability of belonging to the diabetic and non-diabetic class was computed, and the class with the higher probability was selected as the final prediction.

The resulting predictions were used in the next section to evaluate the performance of the classifier and assess its ability to distinguish between diabetic and non-diabetic patients.

3.3 Model Evaluation

To evaluate the performance of the Gaussian Naive Bayes classifier, several classification metrics were computed. The analysis begins with a confusion matrix and continues with accuracy, precision, recall, and F1-score. Together, these measures provide a comprehensive assessment of the model's ability to distinguish between diabetic and non-diabetic patients.

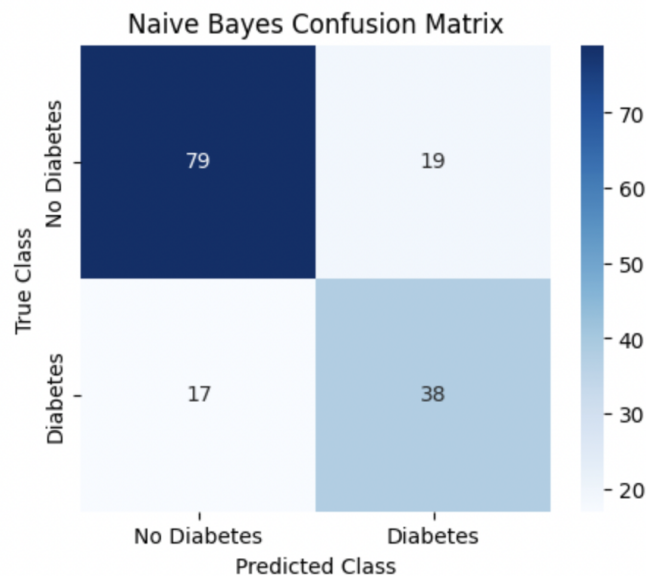


Figure 1: Confusion matrix of the Gaussian Naive Bayes classifier.

Figure 1 summarizes the classification results obtained on the test set. The model correctly classified 79 non-diabetic patients and 38 diabetic patients. At the same time, 19 healthy patients were incorrectly predicted as diabetic, while 17 diabetic patients were classified as non-diabetic.

From a medical perspective, false negative predictions are particularly important because diabetic patients who remain undetected may not receive appropriate treatment and monitoring. However, the relatively balanced number of false positives and false negatives suggests that the classifier does not strongly favor either class.

The overall classification accuracy achieved by the model was 76.47%, indicating that approximately three out of four patients were assigned to the correct class.

In addition to accuracy, precision, recall, and F1-score were calculated:

- Precision = 66.67%
- Recall = 69.09%

- F1-score = 67.86%

The precision value indicates that roughly two-thirds of the patients predicted as diabetic actually belonged to the diabetic class. The recall score shows that the classifier successfully identified nearly 70% of all diabetic patients present in the test set. The F1-score demonstrates a reasonable balance between precision and recall, suggesting that the model provides stable overall performance despite its simplifying assumptions.

Considering the complexity of diabetes prediction and the overlap between the two classes, the obtained results can be considered satisfactory for a baseline classifier.

3.4 Feature Distribution Analysis

To better understand the behavior of the classifier, the distribution of glucose values was examined separately for diabetic and non-diabetic patients. Since glucose is one of the most important indicators of diabetes, analyzing its distribution provides valuable insight into the classification problem.

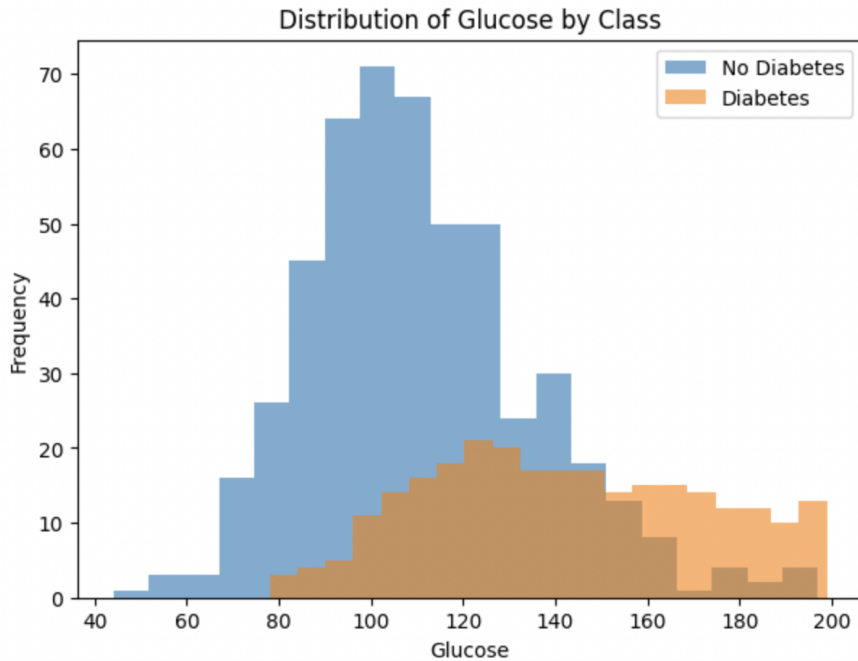


Figure 2: Distribution of glucose values for diabetic and non-diabetic patients.

As shown in Figure 2, diabetic patients generally tend to have higher glucose levels than non-diabetic patients. The distribution of the diabetic class is shifted toward larger values, indicating a strong relationship between elevated glucose levels and the presence of diabetes.

However, a substantial overlap between the two classes can also be observed. In particular, many patients with glucose values between approximately 80 and 140 belong to both classes. As a result, glucose alone is not sufficient to perfectly distinguish diabetic and non-diabetic patients.

This overlap helps explain the classification errors observed in the confusion matrix. Patients located in the overlapping region have similar glucose measurements despite belonging to different classes, making accurate classification more challenging.

3.5 Decision Boundary Visualization

To further investigate the behavior of the Gaussian Naive Bayes classifier, a decision boundary was generated using two selected features: Glucose and BMI. These variables were chosen because they are among the most informative predictors of diabetes and showed strong relationships with the target variable during the exploratory data analysis.

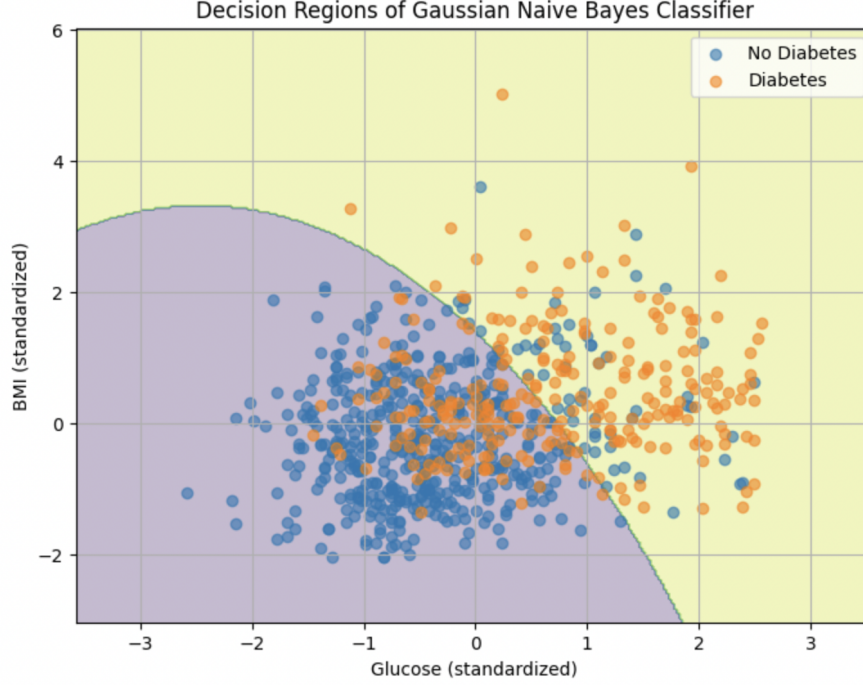


Figure 3: Decision boundary of the Gaussian Naive Bayes classifier using Glucose and BMI.

The decision regions shown in Figure 3 illustrate how the classifier partitions the feature space into diabetic and non-diabetic areas. In general, patients with higher glucose values are more likely to be classified as diabetic, while lower glucose values are associated with the non-diabetic class.

Although a clear trend is visible, the two classes are not perfectly separated. Numerous observations appear near the boundary between the predicted regions, indicating that patients with similar glucose and BMI values may belong to different classes. This observation is consistent with the overlap previously observed in the glucose distribution plot.

Furthermore, diabetes is a complex disease that depends on multiple medical factors rather than only glucose and BMI. While the classifier itself uses all available features, the decision boundary visualization is based on only two variables. Consequently, patients with very similar glucose and BMI values may still belong to different classes because they differ in other characteristics such as age, insulin level, blood pressure, or family history. This explains the presence of overlapping points and highlights the inherent difficulty of diabetes classification.

3.6 Discussion

The Gaussian Naive Bayes classifier achieved an accuracy of 76.47%, with a precision of 66.67%, a recall of 69.09%, and an F1-score of 67.86%. These results indicate that the model is capable of correctly identifying the majority of patients while maintaining a reasonable balance between precision and recall.

The confusion matrix revealed that both false positive and false negative predictions were present. While some diabetic patients were successfully detected, others remained misclassified

due to the overlap between the two classes. This observation was further supported by the glucose distribution analysis and the decision boundary visualization.

Despite its strong independence assumption, Gaussian Naive Bayes successfully captured the main patterns present in the dataset and provided a solid baseline for diabetes prediction. The results obtained in this section will serve as a reference point for comparison with more advanced classification methods in the following sections.

4 Quadratic Discriminant Analysis (QDA)

After evaluating the Gaussian Naive Bayes classifier, we next consider Quadratic Discriminant Analysis (QDA). While both methods are based on Gaussian class distributions, QDA allows each class to have its own covariance matrix. This additional flexibility enables the classifier to model more complex relationships between clinical variables.

The model was implemented from scratch and evaluated on the same training and testing sets used in the previous experiment, allowing a direct comparison with the Gaussian Naive Bayes results.

4.1 Parameter Estimation

Similarly to Gaussian Naive Bayes, the class priors and mean vectors are estimated from the training data. In addition, QDA estimates a separate covariance matrix for each class, allowing the classifier to model class-specific variability.

These estimated parameters form the basis of the multivariate Gaussian distributions used by QDA during classification.

4.2 Multivariate Gaussian Density

Once the parameters have been estimated, QDA models each class using a multivariate Gaussian distribution. The probability density depends not only on the mean values of the features but also on their covariance structure.

Unlike Gaussian Naive Bayes, which treats features independently, the multivariate Gaussian distribution takes into account relationships between variables. This allows the classifier to better represent the underlying structure of the data when feature dependencies are present.

For numerical stability, the logarithm of the probability density is used during computation.

4.3 QDA Prediction

Using the estimated parameters, posterior scores are computed for each class. Test samples are assigned to the class with the higher score, producing the final predictions.

4.4 Model Evaluation

The QDA classifier achieved an accuracy of 79.08% on the test set. This indicates that the majority of patients were classified correctly and suggests that the model is able to distinguish between diabetic and non-diabetic individuals with reasonable success.

To obtain a more detailed assessment of the classifier, the confusion matrix and additional performance metrics are analyzed below.

4.5 Confusion Matrix

The confusion matrix provides a detailed overview of the classification results.

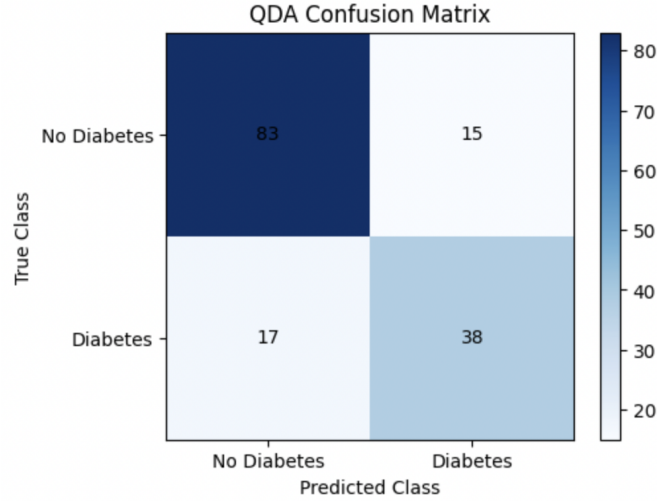


Figure 4: Confusion matrix obtained by the QDA classifier.

As shown in Figure 4, the model correctly classified 83 non-diabetic patients and 38 diabetic patients. In addition, 15 non-diabetic patients were incorrectly classified as diabetic, while 17 diabetic patients were classified as non-diabetic.

The relatively balanced distribution of errors indicates that the classifier does not strongly favor one class over the other. However, the presence of false negatives remains important because diabetic patients may remain undetected.

4.6 Decision Boundary Visualization

To better understand how the QDA classifier separates the two classes, a two-dimensional visualization was created using the Glucose and BMI features.

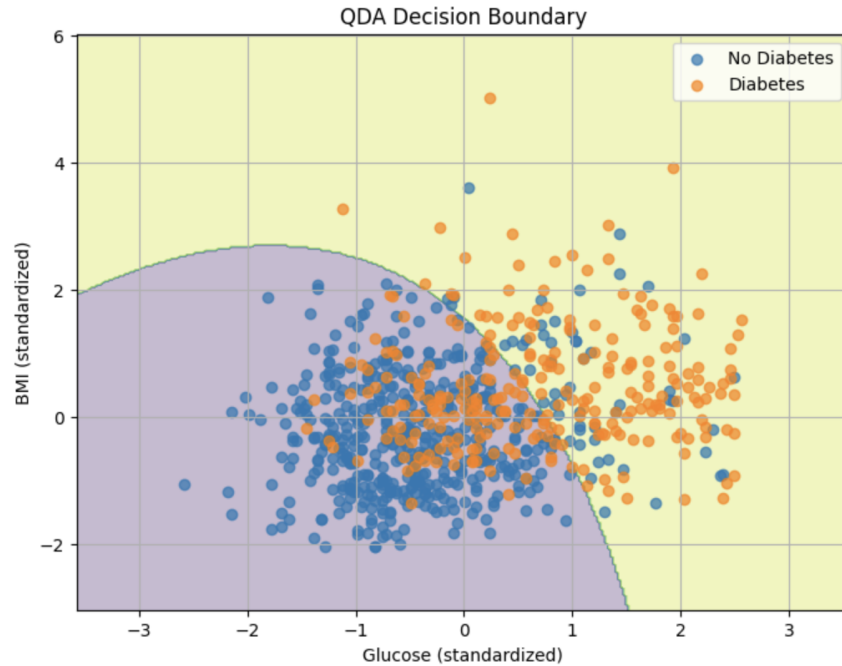


Figure 5: QDA decision boundary using Glucose and BMI features.

Figure 5 illustrates the decision regions learned by the classifier. A clear trend can be

observed, with higher glucose values generally associated with the diabetic class and lower values more frequently associated with the non-diabetic class.

Despite this trend, the two classes are not perfectly separated. A considerable overlap is visible in the central region of the plot, where observations from both classes appear close to the decision boundary. As a result, some patients with similar Glucose and BMI values belong to different classes.

This observation is expected because diabetes is influenced by multiple clinical factors. While Glucose and BMI are among the most informative variables in the dataset, the final classification also depends on other measurements such as Age, Insulin, Blood Pressure, Pregnancies, and Diabetes Pedigree Function. Therefore, a visualization based on only two features cannot completely separate the two classes.

Nevertheless, the figure demonstrates that QDA is able to adapt to the underlying distribution of the data and produce a flexible nonlinear decision boundary.

4.7 Covariance Matrix Analysis

One of the main characteristics of QDA is that it estimates a separate covariance matrix for each class. To better understand the relationships between features, the covariance matrices for diabetic and non-diabetic patients are visualized using heatmaps.

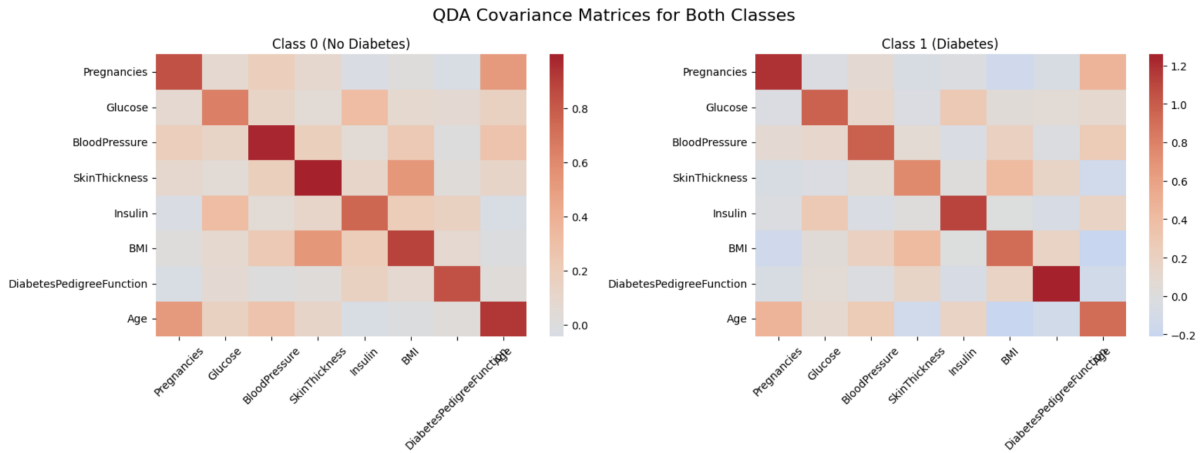


Figure 6: Covariance matrices estimated by QDA for non-diabetic and diabetic patients.

Figure 6 reveals noticeable differences in the covariance structure of the two classes. While some feature relationships are similar, several feature pairs exhibit stronger or weaker covariance depending on the class.

In the diabetic group, variables such as Glucose, BMI, Insulin, and Diabetes Pedigree Function show more pronounced relationships with other clinical measurements. This suggests that the interaction between risk factors is not identical in diabetic and non-diabetic patients.

The covariance heatmaps also indicate that diabetes affects not only the values of individual variables but also the way these variables vary together. Consequently, the two classes exhibit different statistical structures, which helps explain why modeling them separately can improve classification performance.

Overall, the covariance matrices provide valuable insight into the internal structure of the dataset and highlight differences between diabetic and non-diabetic populations.

4.8 Performance Metrics

In addition to accuracy, several evaluation metrics were computed to provide a more comprehensive assessment of the QDA classifier.

- Accuracy: 79.08%
- Precision: 71.70%
- Recall: 69.09%
- F1-score: 70.37%

The obtained results indicate a balanced classification performance across the two classes. The precision value of 71.70% shows that most patients predicted as diabetic were correctly identified. At the same time, the recall of 69.09% demonstrates that the model successfully detected a substantial proportion of diabetic patients.

The F1-score of 70.37% reflects a good compromise between precision and recall, suggesting that the classifier maintains a reasonable balance between detecting diabetic patients and limiting incorrect positive predictions.

Taken together, the accuracy, confusion matrix, and performance metrics indicate that QDA performs reliably on the diabetes dataset and is capable of capturing important patterns within the clinical measurements.

5 K-Nearest Neighbors Classification (KNN)

After evaluating the probabilistic classifiers, we next consider the k-Nearest Neighbors (kNN) algorithm. Unlike Naive Bayes and QDA, kNN does not assume any particular distribution for the data. Instead, it classifies each new observation by identifying the k nearest training samples based on Euclidean distance and assigning the majority class among them. Because no explicit training step is required, kNN is often referred to as a *lazy learner* — all computation is deferred to prediction time.

The model was implemented from scratch and evaluated on the same training and test sets used in the previous experiments, allowing a direct comparison of results.

5.1 Model Implementation

For each test observation, the algorithm computes the Euclidean distance to every training point, selects the k nearest neighbors, and predicts the class by majority vote. The implementation uses only NumPy and does not rely on any external machine learning library.

5.2 Hyperparameter Analysis — Choosing k

The number of neighbors k is the only hyperparameter of the kNN algorithm. A small value of k makes the model sensitive to individual training points and prone to overfitting, while a large k produces a smoother decision boundary but may cause underfitting by including distant and potentially irrelevant neighbors in the vote.

To select the optimal value, classification accuracy was evaluated for all integer values of k from 1 to 30. The value that produced the highest test accuracy was selected for final evaluation.

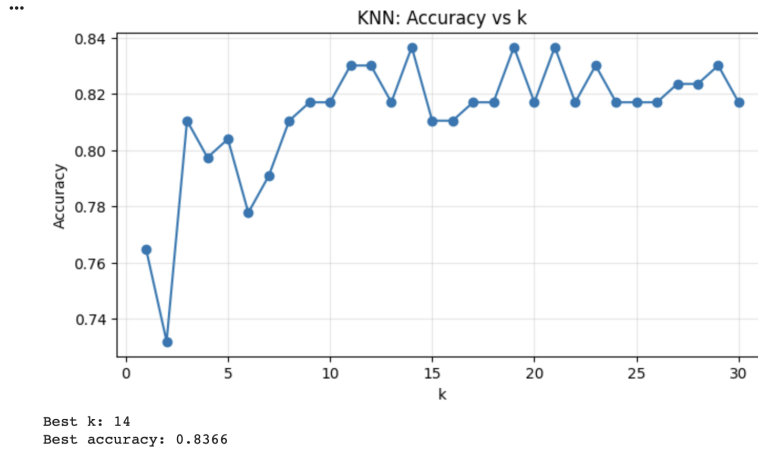


Figure 7: KNN classification accuracy as a function of k .

As shown in Figure 7, the optimal value was found to be $k = 14$, yielding a test accuracy of 83.66%.

5.3 Model Evaluation

The kNN classifier was trained using the standardized training data and evaluated on the 153 test observations. For each test sample, the 14 nearest neighbors were identified and the majority class was assigned as the predicted label.

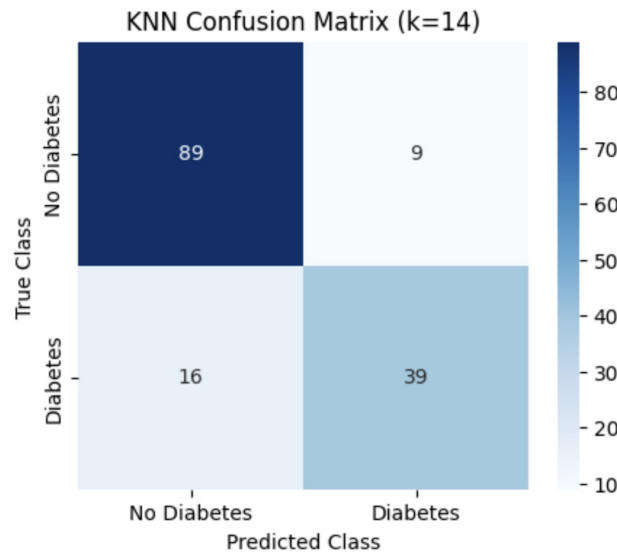


Figure 8: Confusion matrix of the KNN classifier ($k = 14$).

As shown in Figure 8, the model correctly classified 89 non-diabetic patients and 39 diabetic patients. At the same time, 9 non-diabetic patients were incorrectly predicted as diabetic, while 16 diabetic patients were classified as non-diabetic.

As we said before, the presence of false negatives is particularly important in a medical context, as diabetic patients who are not detected may not receive appropriate treatment. The model achieves an accuracy of 83.66%, which represents a clear improvement over both Naive Bayes (76.47%) and QDA (79.08%).

In addition to accuracy, precision, recall, and F1-score were computed:

- Precision = 81.25%

- Recall = 70.91%
- F1-score = 75.73%

The precision of 81.25% indicates that the majority of patients predicted as diabetic were correctly identified. The recall of 70.91% means that approximately three out of ten diabetic patients were missed by the model. The F1-score of 75.73% reflects a reasonable balance between the two metrics.

5.4 Decision Boundary Visualization

To better understand how the kNN classifier separates the two classes, a two-dimensional decision boundary was constructed using the Glucose and BMI features. These variables were selected because they showed the strongest association with the Outcome variable during the exploratory data analysis. Insulin, although more strongly correlated with Outcome, was excluded due to its high number of outliers, which would distort the visualization. The actual classifier uses all eight features; the figure represents a two-dimensional approximation only.

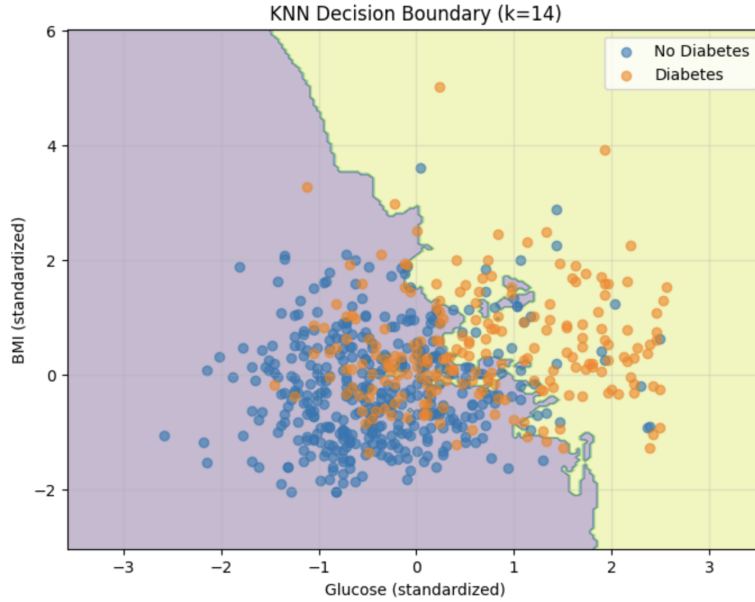


Figure 9: KNN decision boundary using Glucose and BMI features ($k = 14$).

Unlike Naive Bayes and QDA, kNN produces an irregular, non-smooth decision boundary that adapts to the local structure of the training data. A considerable overlap between the two classes is visible in the central region, which is consistent with the classification errors observed in the confusion matrix.

6 Distance-Weighted KNN

Distance-Weighted kNN is an established extension of the standard kNN algorithm in which closer neighbors are assigned greater influence over the prediction than more distant ones [1].

In plain kNN, every neighbor contributes one equal vote. In Distance-Weighted kNN, each neighbor contributes a vote weighted by

$$w_i = \frac{1}{d_i + \varepsilon},$$

where d_i is the Euclidean distance from the query point to the i -th neighbor, and ε is a small constant added to prevent division by zero in the rare case where a training point coincides exactly with the query point.

This weighting scheme is intuitively justified: a neighbor that is very close to the query point is more likely to share the same class, and should therefore carry more weight in the vote than one that is far away and may belong to a different local region of the feature space.

The model was implemented from scratch and evaluated on the same data splits used in the previous sections.

6.1 Implementation

The implementation differs from plain kNN only in the voting step. After identifying the k nearest neighbors, each neighbor's contribution is multiplied by $1/(d_i + \varepsilon)$ rather than counted as a unit vote. The class with the highest total weighted score is assigned as the predicted label.

6.2 Hyperparameter Analysis — Choosing k

As with plain kNN, the optimal value of k was determined by evaluating classification accuracy for all integer values from 1 to 30.

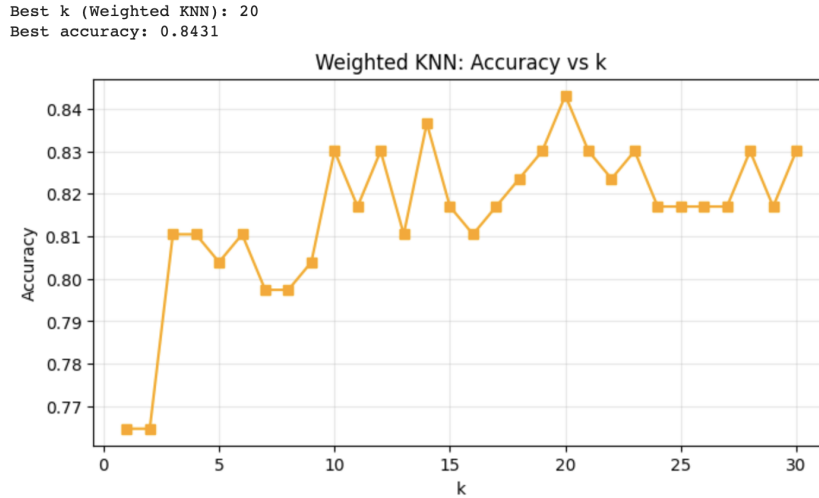


Figure 10: Distance-Weighted KNN classification accuracy as a function of k .

The optimal value was found to be $k = 20$, which is notably larger than the optimal $k = 14$ found for plain kNN. This is consistent with the motivation for distance weighting: because distant neighbors are down-weighted, the model can consider a larger neighborhood without being misled by irrelevant points.

6.3 Model Evaluation

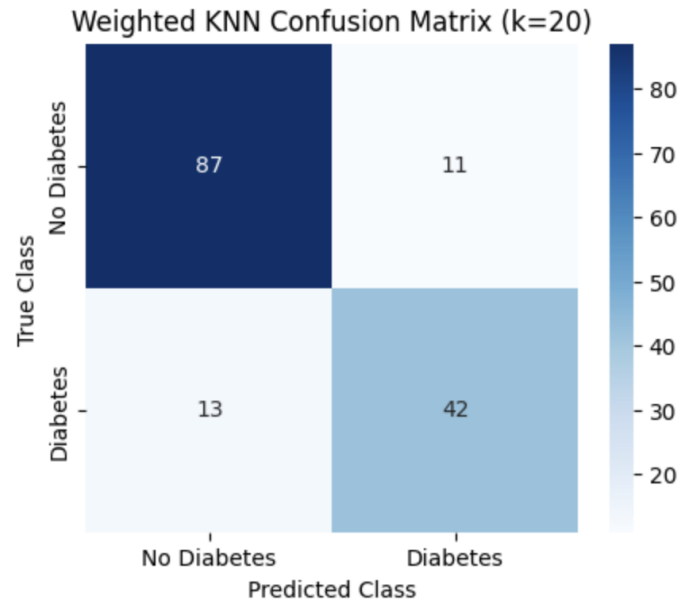


Figure 11: Confusion matrix of the Distance-Weighted KNN classifier ($k = 20$).

As shown in Figure 11, the model correctly classified 87 non-diabetic patients and 42 diabetic patients. Compared to plain kNN, the number of false negatives decreased from 16 to 13, meaning that fewer diabetic patients were missed.

The performance metrics confirm this improvement:

- Accuracy = 84.31%
- Precision = 79.25%
- Recall = 76.36%
- F1-score = 77.78%

The recall improved from 70.91% to 76.36%, indicating that the model now correctly identifies a larger proportion of diabetic patients. This improvement is particularly relevant from a medical perspective, as failing to detect diabetes can have serious consequences for the patient. Distance-Weighted kNN outperforms plain kNN across all evaluation metrics, demonstrating that assigning greater influence to closer neighbors leads to a more accurate and reliable classifier.

7 Feature-Weighted KNN

In the previous kNN variants, all features contributed equally to the distance calculation. However, as shown in the exploratory data analysis, the features in this dataset differ significantly in their relevance to the classification task — Glucose, Insulin, and BMI are strongly correlated with the Outcome, while BloodPressure and DiabetesPedigreeFunction show considerably weaker associations.

Treating all features equally therefore means that less informative measurements contribute as much to the distance calculation as the most relevant clinical indicators. This motivated the introduction of Feature-Weighted kNN, where each feature is assigned a weight proportional to

its correlation with the target variable. In this way, the model focuses on the most diagnostically relevant measurements when identifying the nearest neighbors, rather than treating all clinical variables as equally important.

In standard kNN all features contribute equally to the distance calculation. Feature-Weighted kNN addresses this limitation by assigning each feature a weight that reflects its relevance to the classification task [2]. The weighted Euclidean distance between two observations $\mathbf{x}$ and $\mathbf{x}'$ is defined as

$$d(\mathbf{x}, \mathbf{x}') = \sqrt{\sum_{j=1}^p w_j (x_j - x'_j)^2},$$

where $w_j \geq 0$ is the weight assigned to feature j .

This approach differs from Distance-Weighted kNN: here the weights are assigned to *features* rather than to neighbors. The model was implemented from scratch and evaluated on the same data splits used in the previous sections.

7.1 Feature Weights

The feature weights are derived from the Pearson correlations with the target variable Outcome computed on the training data. Features with a stronger correlation with the target are assigned a higher weight, so they contribute more to the distance calculation. The weights are normalized to sum to one to ensure comparability with the unweighted distance.

Table 1: Feature weights derived from Pearson correlation with Outcome

Feature	Weight
Glucose	0.217
Insulin	0.165
BMI	0.137
SkinThickness	0.130
Age	0.104
Pregnancies	0.097
BloodPressure	0.076
DiabetesPedigreeFunction	0.074

7.2 Implementation

The only modification with respect to plain kNN is that the squared difference between each feature is multiplied by its corresponding weight before the Euclidean distance is computed. Features with higher weights, such as Glucose and Insulin, therefore exert greater influence on which training samples are considered neighbors of a given query point.

7.3 Hyperparameter Analysis — Choosing k

The optimal value of k was determined by evaluating accuracy for all integer values from 1 to 30.

Best k (Feature-Weighted KNN): 3
 Best accuracy: 0.8431

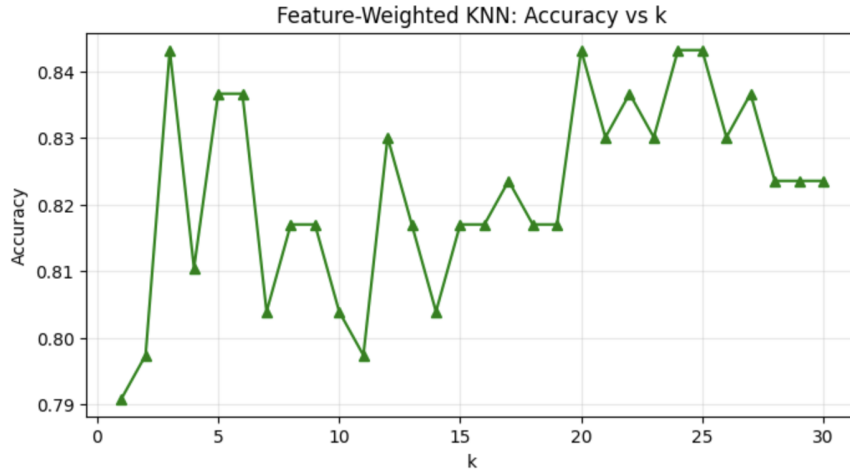


Figure 12: Feature-Weighted KNN classification accuracy as a function of k .

The optimal value was found to be $k = 3$, which is notably smaller than the values obtained for plain kNN ($k = 14$) and Distance-Weighted kNN ($k = 20$). By assigning greater weight to the most informative features, the model can rely on a smaller and more relevant neighborhood to make accurate predictions, reducing the need for a large number of neighbors.

7.4 Model Evaluation

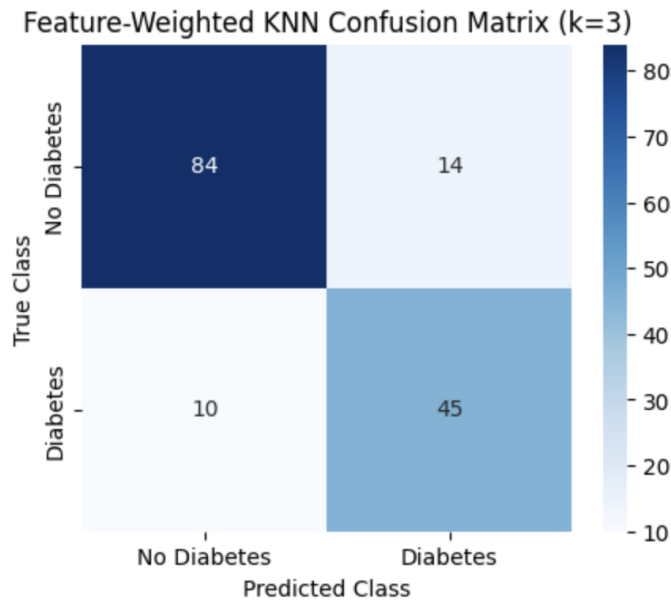


Figure 13: Confusion matrix of the Feature-Weighted KNN classifier ($k = 3$).

As shown in Figure 13, the model correctly classified 84 non-diabetic patients and 45 diabetic patients, with only 10 false negatives. The performance metrics are as follows:

- Accuracy = 84.31%
- Precision = 76.27%

- Recall = 81.82%
- F1-score = 78.95%

Although Feature-Weighted kNN achieves the same overall accuracy as Distance-Weighted kNN, it outperforms it in recall and F1-score. The recall of 81.82% is the highest among all three kNN variants, meaning that the model successfully identified more than eight out of ten diabetic patients. In a medical context, this makes Feature-Weighted kNN the preferred choice, as failing to detect diabetes is generally more harmful than a false positive prediction.

8 Results, Analysis, and Conclusions

8.1 Comparative Summary of Results

This section presents a consolidated comparison of all five classification methods evaluated in this project: Gaussian Naive Bayes, Quadratic Discriminant Analysis (QDA), k-Nearest Neighbors (kNN), Distance-Weighted kNN, and Feature-Weighted kNN. All models were trained and evaluated on the same data splits and under identical preprocessing conditions, ensuring a fair and direct comparison.

Table 2 summarizes the performance of each classifier on the test set of 153 patients.

Table 2: Classification performance of all five methods on the test set

Method	Accuracy	Precision	Recall	F1-score
Gaussian Naive Bayes	76.47%	66.67%	69.09%	67.86%
QDA	79.08%	71.70%	69.09%	70.37%
kNN ($k = 14$)	83.66%	81.25%	70.91%	75.73%
Distance-Weighted kNN ($k = 20$)	84.31%	79.25%	76.36%	77.78%
Feature-Weighted kNN ($k = 3$)	84.31%	76.27%	81.82%	78.95%

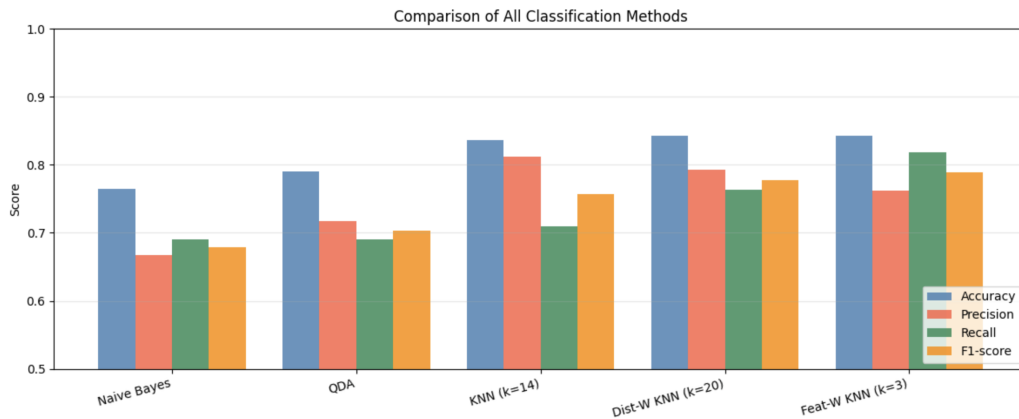


Figure 14: Comparison of all five classifiers across evaluation metrics.

8.2 Analysis of Results

8.2.1 Gaussian Naive Bayes

Gaussian Naive Bayes achieved the lowest performance among all five methods, with an accuracy of 76.47% and an F1-score of 67.86%. This outcome is consistent with the limitations identified during the exploratory data analysis. The Naive Bayes classifier assumes that all features are

conditionally independent given the class, and that each feature follows a Gaussian distribution within each class.

Both assumptions were found to be only partially satisfied in this dataset. The correlation analysis revealed moderate inter-feature correlations, particularly between Age and Pregnancies (0.54) and between SkinThickness and BMI (0.57), which violates the independence assumption. Furthermore, several features — including Insulin, Age, and Pregnancies — exhibit right-skewed distributions that deviate from the assumed Gaussian shape.

Despite these violations, Naive Bayes still managed to capture the main predictive patterns in the data and provided a useful baseline for comparison. Its main advantage remains its simplicity and computational efficiency.

8.2.2 Quadratic Discriminant Analysis

QDA outperformed Gaussian Naive Bayes on all metrics, achieving an accuracy of 79.08% and an F1-score of 70.37%. Unlike Naive Bayes, QDA takes into account the full covariance structure of each class, which allows it to model the relationships between features rather than treating them as independent. The covariance matrix analysis confirmed that the two classes exhibit noticeably different statistical structures, which QDA is able to exploit.

Both Naive Bayes and QDA share the same recall of 69.09%, meaning that both models miss the same proportion of diabetic patients. The improvement of QDA over Naive Bayes is primarily reflected in higher precision and accuracy, suggesting that QDA makes fewer false positive errors.

Both methods belong to the family of probabilistic classifiers based on the Bayes decision rule, and both assume Gaussian class-conditional distributions. The key difference is that QDA relaxes the independence assumption of Naive Bayes by modeling the full multivariate Gaussian distribution with a separate covariance matrix per class.

8.2.3 k-Nearest Neighbors

kNN achieved a substantially higher accuracy of 83.66% compared to both probabilistic classifiers, demonstrating that a non-parametric approach can better capture the underlying structure of this dataset. Unlike Naive Bayes and QDA, kNN makes no assumptions about the distribution of the data and instead relies entirely on the local geometry of the feature space.

The recall of 70.91% is similar to that of the probabilistic methods, indicating that kNN still misses approximately three out of ten diabetic patients. However, the precision of 81.25% is notably higher than that of Naive Bayes (66.67%) and QDA (71.70%), meaning that kNN produces fewer false positive predictions.

The optimal number of neighbors was found to be $k = 14$, reflecting a moderate balance between sensitivity to local structure and robustness to noise.

8.2.4 Distance-Weighted kNN

Distance-Weighted kNN improved upon plain kNN across all evaluation metrics. By assigning greater influence to closer neighbors, the model achieves a higher recall of 76.36% compared to 70.91% for plain kNN, meaning that fewer diabetic patients are missed. The accuracy increased from 83.66% to 84.31%, and the F1-score improved from 75.73% to 77.78%.

The optimal value of $k = 20$ is larger than the $k = 14$ found for plain kNN, which confirms the theoretical motivation for distance weighting: because distant neighbors are down-weighted, the model can safely consider a larger neighborhood without being misled by irrelevant training samples.

8.2.5 Feature-Weighted kNN

Feature-Weighted kNN achieved the same overall accuracy as Distance-Weighted kNN (84.31%), but outperformed it in recall (81.82% vs. 76.36%) and F1-score (78.95% vs. 77.78%). This is the highest recall among all five methods, meaning that Feature-Weighted kNN correctly identified the largest proportion of diabetic patients in the test set.

The motivation for this method arose directly from the exploratory data analysis: the features in this dataset differ considerably in their correlation with the target variable. Treating all features equally in the distance calculation is therefore suboptimal. By weighting each feature according to its Pearson correlation with the Outcome variable, the model focuses on the most diagnostically relevant clinical indicators — particularly Glucose and Insulin — when identifying the nearest neighbors.

The optimal number of neighbors was only $k = 3$, which is notably smaller than the values found for the other kNN variants. This suggests that, once the feature space is properly weighted, even a very small neighborhood provides sufficient information for accurate classification.

8.3 Key Findings

Based on the comparative evaluation, the following conclusions can be drawn:

- **Non-parametric methods outperform probabilistic classifiers** on this dataset. Both kNN variants achieved accuracy above 83%, while Naive Bayes and QDA remained below 80%. This suggests that the distributional assumptions of the probabilistic methods are not fully satisfied by the data, and that the class boundaries are better captured by local, distance-based approaches.
- **Feature weighting provides the best recall.** Feature-Weighted kNN achieved the highest recall of 81.82%, correctly identifying more than eight out of ten diabetic patients. In a medical context, recall is the most critical metric, as failing to detect diabetes can have serious consequences for the patient. For this reason, Feature-Weighted kNN is considered the best-performing method in this study.
- **Distance weighting improves upon plain kNN.** Assigning greater influence to closer neighbors consistently improved all metrics compared to unweighted kNN, confirming the theoretical motivation for this extension.
- **The independence assumption of Naive Bayes is a limiting factor.** The moderate inter-feature correlations observed in the EDA (e.g. SkinThickness–BMI: 0.57, Age–Pregnancies: 0.54) partially violate this assumption, which contributes to the lower performance of Naive Bayes relative to QDA and kNN.
- **QDA captures class-specific covariance structure.** The covariance matrix analysis confirmed that the two classes exhibit different statistical structures. QDA exploits this by estimating a separate covariance matrix per class, which explains its improvement over Naive Bayes despite sharing the same Gaussian assumption.

8.4 Limitations

Several limitations of this study should be noted. The dataset contains only 768 observations, which limits the statistical reliability of the results. The class imbalance (65% non-diabetic, 35% diabetic) may affect model performance, and no resampling techniques were applied to address it. The evaluation was performed on a single train-test split; cross-validation would provide more robust estimates of generalization performance.

8.5 Future Work

Several directions could further improve the results obtained in this study. Cross-validation could be applied to obtain more reliable performance estimates and reduce the dependence on a single data split. Resampling strategies such as SMOTE could be used to address the class imbalance and potentially improve recall across all methods.

From a methodological perspective, the two kNN extensions evaluated in this project — distance weighting and feature weighting — could be combined into a single model that applies both types of weighting simultaneously. Support Vector Machines (SVM) or ensemble methods such as Random Forests could also be explored as additional baselines to assess how more complex models compare to the methods studied here.

Finally, more sophisticated feature weighting strategies, such as optimization-based or mutual-information-based weighting, could be explored to further improve the performance of Feature-Weighted kNN.

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